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Palaeomagnetism of the upper zone of the Bushveld Complex

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Abstract

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Palaeomagnetic data were acquired from eight sampling sites situated in the upper zone of the Bushveld Complex, Transvaal, South Africa. Specimens were subjected to alternating field and thermal demagnetization. A mean magnetization direction, yielding a palaeomagnetic pole at 16.1° S, 148.5° W, was found. Fold tests indicate that the upper zone had acquired its remanent magnetization with the igneous layering in its present position. With this new data on the layered sequence, it is possible to divide the mafic portion into three magnetic polarity zones, namely MPZA, MPZB and MPZC.

Introduction

The first palaeomagnetic investigation on the Bushveld Complex were done by Gough and Van Niekerk (1959). They presented data from five sampling sites, which yielded a mean palaeomagnetic pole position at 23° N, 36° E. The rock units which they sampled in the western and eastern Bushveld Complex are now formally known as the Pyramid Gabbro-Norite and the Leolo Mountain Gabbro-Norite (SACS, 1980) or informally as the main zone (Fig. 1). A more detailed investigation covering the critical, main and upper zones was initiated by the author in 1980. Results from the critical and main zones have been reported elsewhere (Hattingh, 1986a, 1986b, 1986c).

Eight sampling sites were selected in the upper zone (Fig. 1). Of these, three sites are in quarries, two in road cuttings and the remainder are situated in streambeds. The distribution and the relatively few sampling sites, reflect the general absence of suitable sampling sites in the upper zone.

Samples were taken with a petrol driven portable core drill. The core samples are 25 mm in diameter and up to 200 mm in length. Four to five core samples were taken at each sampling site. Each core was cut into an average of five cylindrical specimens. Core orientation was done with a sun compass (Creer and Sanver, 1967) which allows orientation to a precision of 1° (Gough, 1967).

Laboratory measurements

The remanent magnetization of specimens was measured using a Digico "Complete Results" fluxgate spinner magnetometer. In practice it was found that to measure a reliable direction of magnetization to within one degree, a minimum intensity of magnetization of 0.5×10^{-3} A m^{-1} was required.

Continuous thermal demagnetization of specimens was done with a slightly-modified form of the spinner magnetometer. It was possible to measure remanence at high temperatures without the

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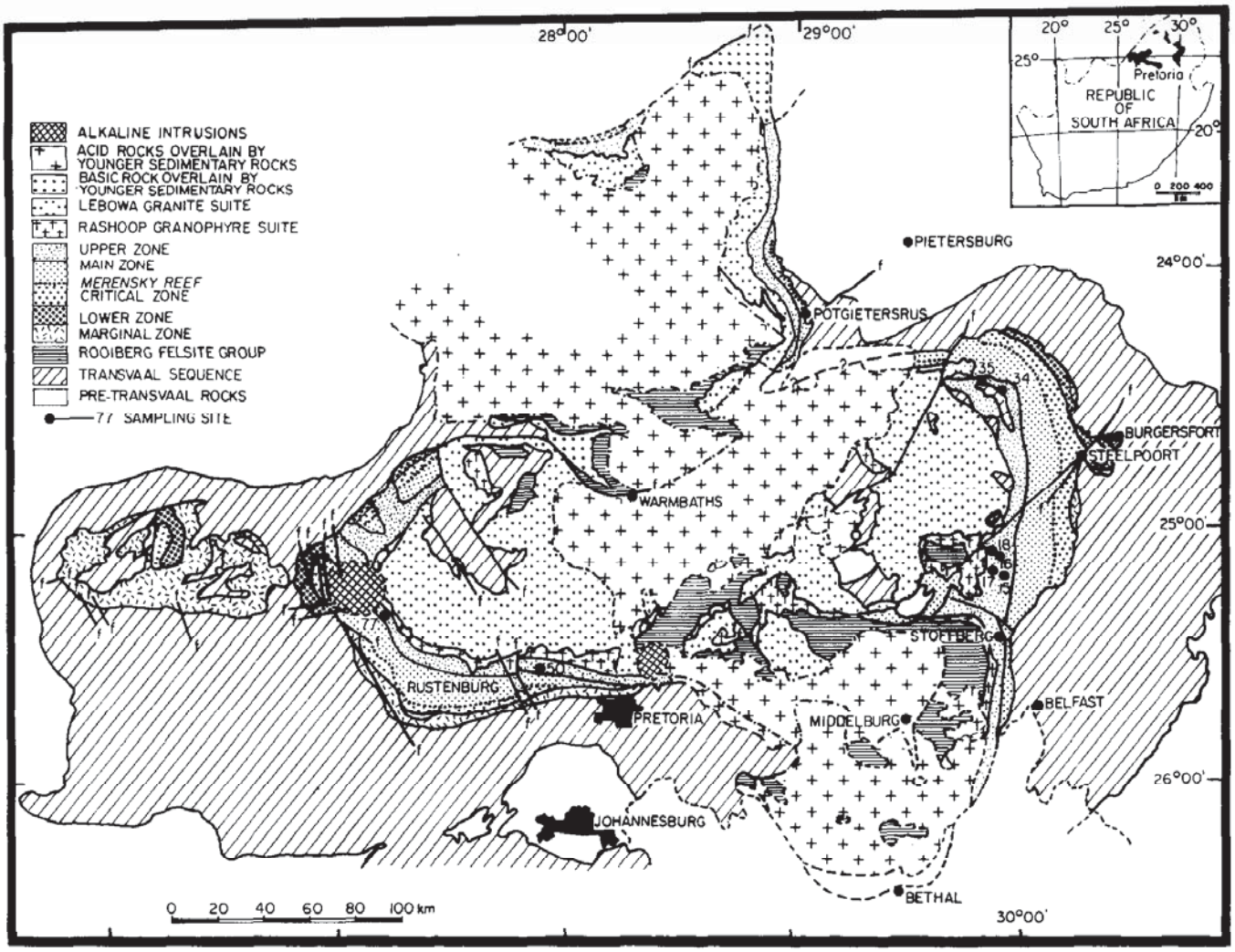


Fig. 1. Geological map of the Bushveld Complex showing sampling sites in the upper zone.

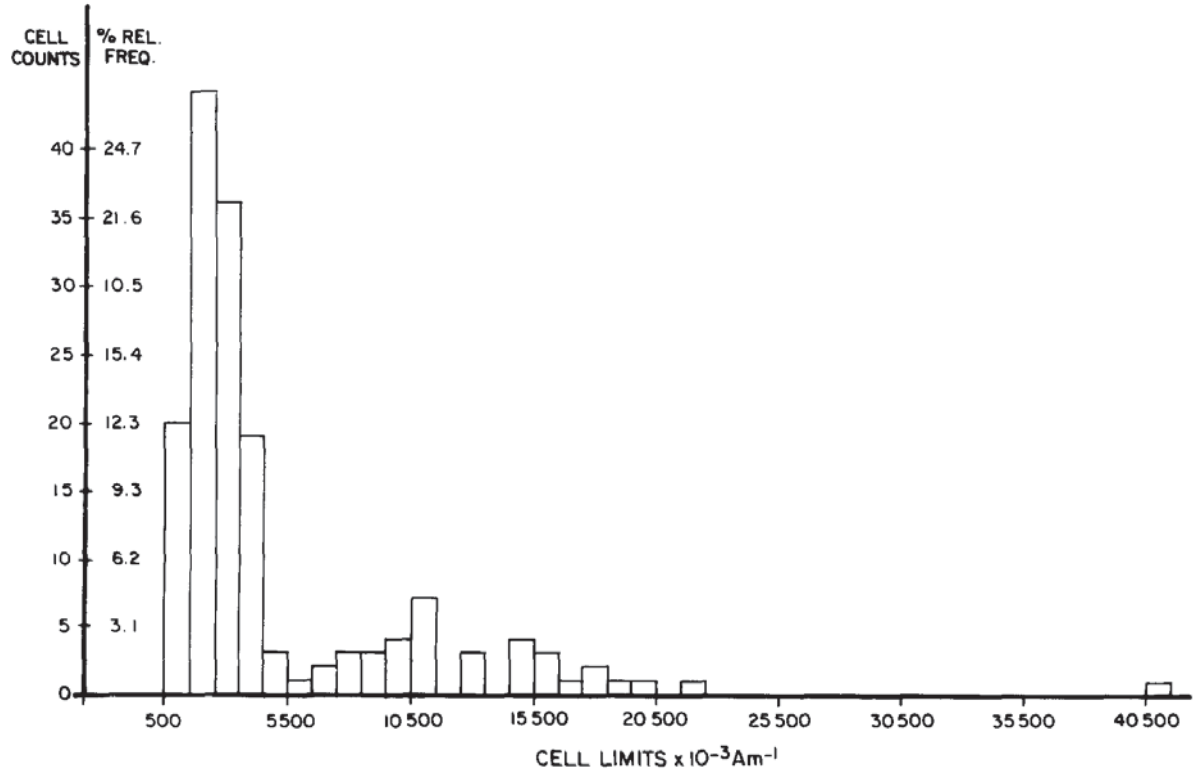


Fig. 2. Histogram plot of the NRM intensities of specimens from the upper zone. Cell width is 1000×10^{-3} A m^{-1}.

TABLE 1

Natural remanent magnetization directions of sites in the upper zone of the Bushveld Complex *

Site	N	D (°)	I (°)	α_{95} (°)	k
15	4	264.5	−74.4	35.0	7
17	4	335.5	−3.4	11.9	60
18	4	44.8	−18.9	22.9	17
34	4	289.2	−11.9	46.0	5
35	4	223.2	−4.4	40.6	10
50	5	145.2	−65.4	9.4	67
77	4	158.6	−47.5	22.4	18

* N is the number of core samples; D is the declination of the magnetization direction, measured eastwards from north; I is the inclination of the magnetization direction, downwards positive, upwards negative; α_{95} is the semi-angle of cone of 95% confidence; k is the estimate of the precision parameter κ.

need for cooling the specimen in a field free space. Operating the instrument in this mode, however, led to a decreased sensitivity, and it was found that the minimum intensity of magnetization that could be measured reliably was in the region of 4×10^{-3} A m^{-1}.

Alternating field (AF) demagnetization of specimens was done with an apparatus built at the Geological Survey of South Africa following a description by McElhinny (1964). This apparatus is capable of creating a peak alternating magnetic field of 100 mT.

The average intensity of the NRM of specimens from the upper zone is 6070×10^{-3} A m^{-1}. Although this is lower than that of specimens from the main zone (Hattingh, 1986b, 1986c), the range of intensities is remarkably broad and specimens with intensities as high as $41{,}000 \times 10^{-3}$ A m^{-1} and as low as 800×10^{-3} A m^{-1} were measured (Fig. 2).

The NRM directions of seven sites were consistent at sample level, but there is no consistency of directions between sites. In Table 1 the mean NRM directions of these sites with the relevant statistics for each site are listed, and these NRM directions are stereographically presented in Fig. 3a.

Data from one site (site 16) were rejected, because of a random distribution (95 per cent

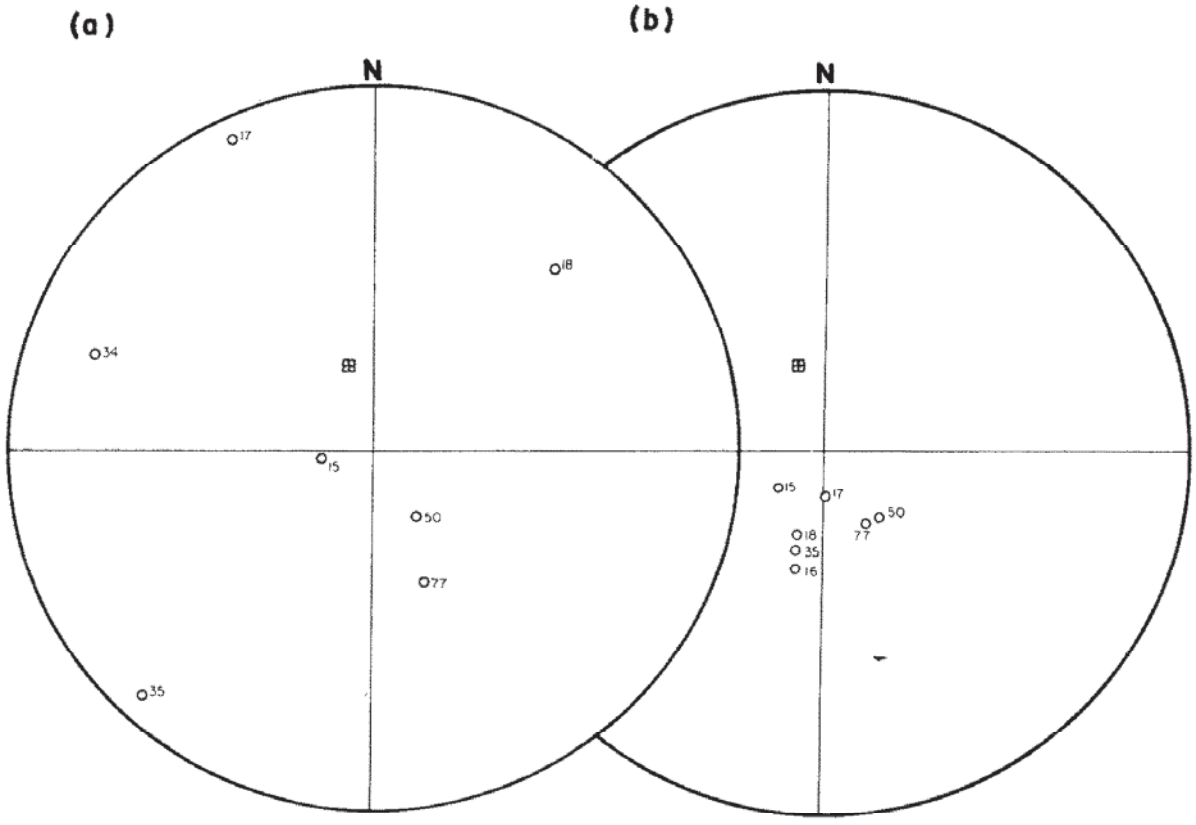


Fig. 3. a. Equal angle projections of NRM directions of sites. b. Magnetization directions of sites after AF demagnetization. ○ = upper hemisphere. Present-day magnetic field direction upward indicated by ⊞.

confidence limit, Watson, 1956) of directions from specimens. The scatter of mean site NRM directions (Fig. 3) is random and cannot be grouped together.

Alternating field demagnetization

Site mean magnetization directions obtained after bulk AF demagnetization at an optimum AF value indicated by the Briden SI (Briden, 1972), are listed in Table 2.

During stepwise AF demagnetization the change in the magnetization directions from one site (site 34) was random and the site was rejected from further analyses on the grounds of an unstable magnetization. Table 2 further contains one extra site (site 16), which shows a significant grouping of magnetization directions of specimens only after bulk AF demagnetization.

The magnetization directions of sites after bulk AF demagnetization (Fig. 3b), form group BUZ1AF with $N = 7$, $D = 184.3°$; $I = -66.0$; $\alpha_{95} = 11.0°$, $k = 31$. If a fold test (Graham, 1949) is applied to the magnetization directions obtained by bulk AF demagnetization (Table 2) the corrected directions from group BUZ1AFR with $N = 7$; $D = 175.1°$; $I = -60.9°$; $\alpha_{95} = 10.6°$; $k = 33$. The improvement in k from 31 to 33 after rotation is not significant and this result is at variance with the results obtained from the main and critical zones (Hattingh, 1986a, 1986b, 1986c). Dip directions of the igneous layering at sites used in this test vary considerably and should have led to a significant improvement in k, even though the dip values and dip directions may not be very accurate. Unless the Briden SI (Briden, 1972), has failed completely to locate a primary magnetization component at each site, one has to conclude that the magnetization of the upper zone has been acquired with the igneous layering of the upper zone in its present position.

The circular standard deviation (Δ) of group BUZ1AF is 14.5° with upper and lower limits of 23.9° and 10.9° at the 95% confidence limit according to the method of Cox (1969a). The expected value of Δ for a palaeolatitude of 49° is 11.5° (Brock, 1971). This indicates that the observed scatter of directions in group BUZ1AF, can possibly be due to palaeosecular variation, but it is noteworthy that the value of Δ for group BUZ1AF is appreciably higher than the expected value, which could suggest that there may be other factors influencing the observed scatter of directions. The circular standard deviation of group BUZ1AFR, is not much different ($\Delta = 14.1°$) and the above also holds true for the scatter of this group.

The directional response of specimens from typical sites to stepwise AF demagnetization, is shown in Fig. 4a. During stepwise AF demagneti-

TABLE 2

Magnetization directions of sites in the upper zone of the Bushveld Complex after bulk alternating field demagnetization (A), (B) with the igneous layering in a horizontal position *, and (C) the dip and dip directions ** of the igneous layering at each site

Site	Alternating field (mT)	A				B		C	
		D (°)	I (°)	α_{95} (°)	k	D (°)	I (°)	dip (°)	dip direction (°)
15	40	231.1	−71.0	9.1	101	168.8	−75.8	18	N277
16	40	194.8	−52.4	17.9	27	174.5	−46.6	18	N293
17	20	180.5	−75.1	23.9	16	165.2	−66.5	10	N321
18	40	198.3	−62.9	6.1	224	178.2	−64.4	10	N270
35	20	196.5	−57.8	6.7	184	179.9	−70.7	15	N220
50	10	141.5	−62.8	7.7	98	155.6	−50.8	15	N06
77	50	150.4	−63.8	35.0	13	197.6	−46.0	35	N57

* D is the declination of the magnetization direction; measured eastwards from north; I is the inclination of the magnetization direction; downwards positive, upwards negative; α_{95} is the semi-angle cone of 95% confidence; k is the estimate of the precision parameter κ.

** Measured from north in a clockwise direction.

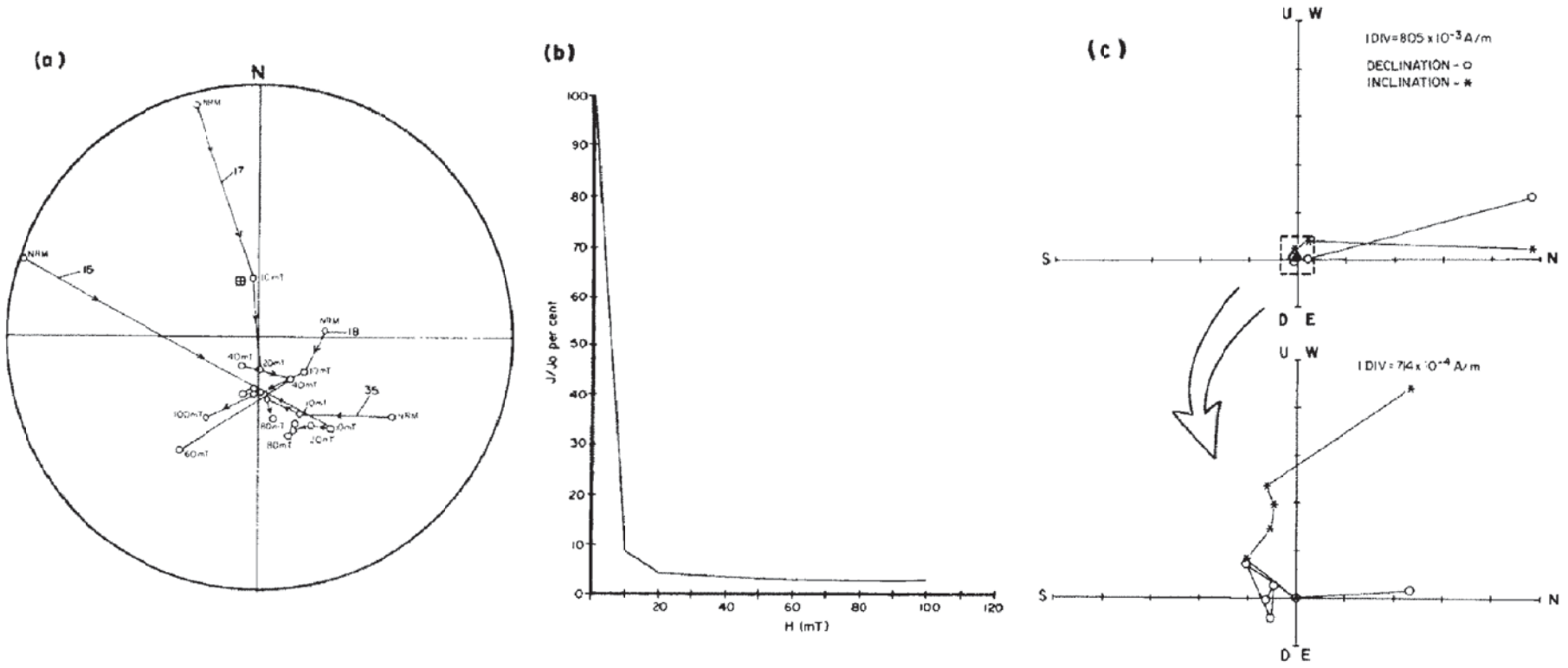


Fig. 4. a. Stepwise AF demagnetization of specimens from sites 16, 17, 18 and 35. Plotting convention as in Fig. 3. b. Typical normalized intensity response curve to AF demagnetization. Specimen from site 17. c. Orthogonal plot of AF demagnetization response, specimen from site 17.

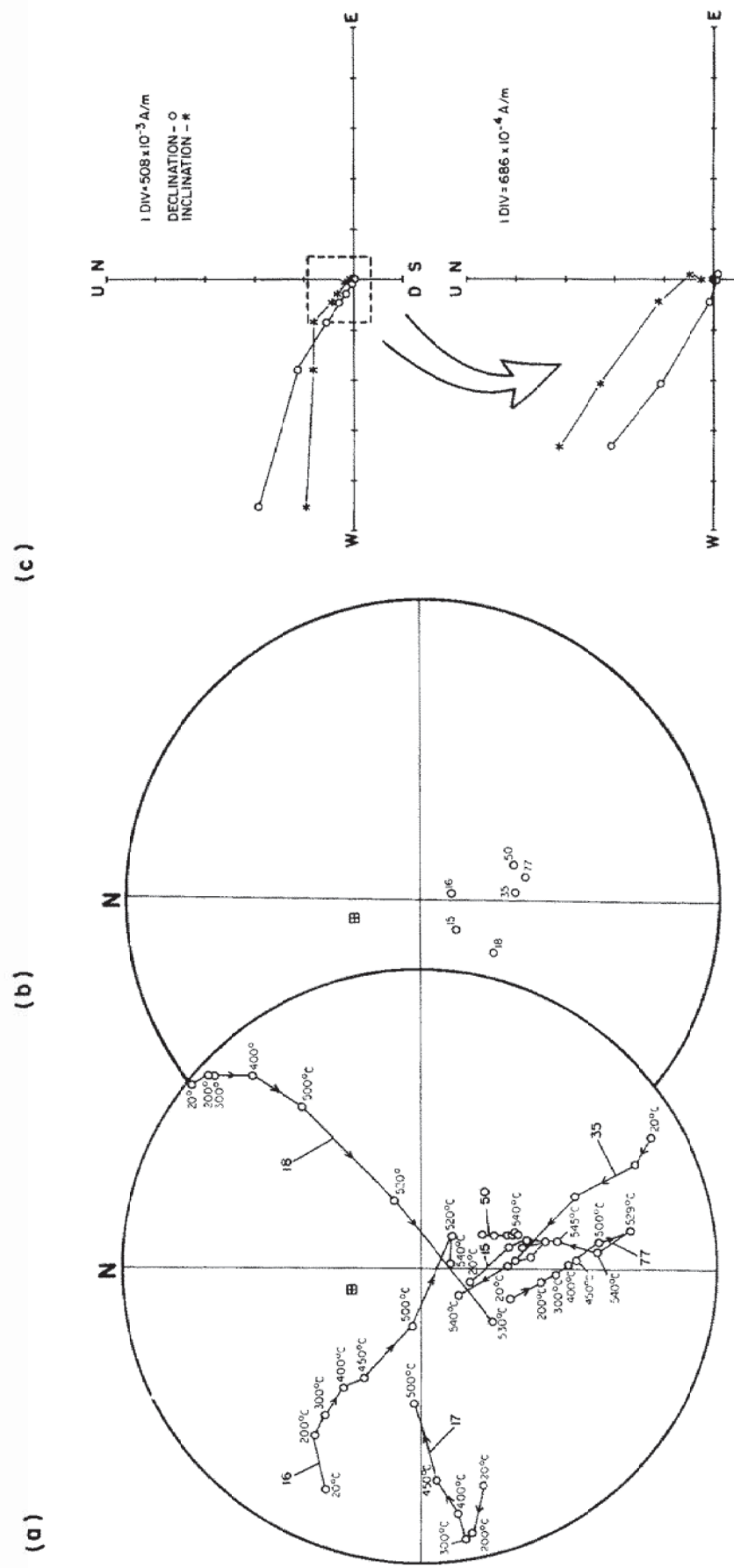


Fig. 5. a. Directional response of magnetization vectors of specimens from six sites to continuous thermal demagnetization. b. End point vectors (see text) after continuous thermal demagnetization of specimens. Plotting convention as in Fig. 3. c. Orthogonal plot of thermal demagnetization response, specimen from site 16.

zation the magnetization vectors converge rapidly to the mean direction of group BUZ1AF. At intermediate AF step values, the vectors then move around in the convergence area indicating a stable midpoint, before moving away at high AF values. It is clear from this that the Briden SI (Briden, 1972) succeeded in locating this stable mid-point resulting in the formation of group BUZ1AF after bulk demagnetization of specimens.

The intensity response curves to stepwise AF demagnetization of specimens from the upper zone all have a very characteristic shape, exhibited by no other specimens from the rest of the layered sequence. Such a typical curve, from specimen 17/2D, is shown in Fig. 4b. The curves all indicate that 80–90% of the intensity of the NRM is destroyed in the first 10 mT step during stepwise AF demagnetization. At progressively higher AF values the intensity decreases very slowly and often shows a marginal increase at certain AF steps. This pattern indicates a very "soft", and as seen in Fig. 4, unstable magnetization, which dominates the NRM characteristics of these specimens.

Thermal demagnetization

The directional response to continuous thermal demagnetization of one specimen from each site in the upper zone is shown in Fig. 5a. With increasing temperature all the magnetization vectors converge. For five specimens the convergence paths of the vectors describe parts of circles of remagnetization, while in two instances the converging paths are made up of two segments of remagnetization circles (Fig. 5a). The magnetization vector of only one specimen, from site 50, remained fairly stable during thermal demagnetization, with very little movement initially and then reaching a stable end point. Magnetization directions of specimens from site 34, which had previously been found to become unstable during AF demagnetization, repeated this behaviour during continuous thermal demagnetization and all data from this site were rejected.

The magnetization vectors shown in Fig. 5a are all resultant vectors of two or more magnetization vectors. At very high temperatures, before reaching the Curie point of the specimens, the convergence is a maximum, indicating that most, if not all the secondary magnetization components have been removed by the thermal demagnetization. If these end point vectors (magnetization directions measured just before reaching the Curie point, or in some cases, the last magnetization which could be measured accurately at high temperatures) of the specimens were grouped together, the mean group direction should yield a value the same as or close to the direction of the primary magnetization of the specimens. However, it is clear from Fig. 5, that the magnetization direction from site 17 has not fully converged at 500 °C, where the last accurate measurement of the magnetization vector was made and consequenlty, this site will have to be rejected from a group comprising these end point vectors. The end point vectors are shown in Fig. 5b and constitute group BUZ2T with $N = 6$; $D = 182.6°$; $I = -62.9°$; $\alpha_{95} = 13.7$; $k = 25$.

Group BUZ2T does not differ much from group BUZ1AF, which was obtained after bulk AF demagnetization; in fact, the respective circles of 95% confidence overlap. This suggests that both the AF demagnetization and the thermal demagnetization succeeded in approximately locating the primary magnetization of the upper zone.

Because the continuous thermal demagnetization of a specimen is completed without removing the specimen from the field free space of the spinner magnetometer, one would expect adverse effects such as the acquisition of an ARM or a VRM by the specimens during demagnetization, to be absent. The magnetization directions of group BUZ2T should thus be more suitable for a fold test than those directions obtained from AF demagnetization of specimens. In Fig. 6a the end point vectors are shown after correction for the dip of the igneous layering at each site. This yields group BUZ2TR with $N = 6$; $D = 172.6°$; $I = -57.3°$; $\alpha_{95} = 18°$; $k = 15$.

The value of the estimate k of the precision parameter, has decreased from 25 to 15. This indicates a weaker grouping of directions with the igneous layering in a horizontal position, which suggests that the primary magnetization of the upper zone was acquired with the igneous layering

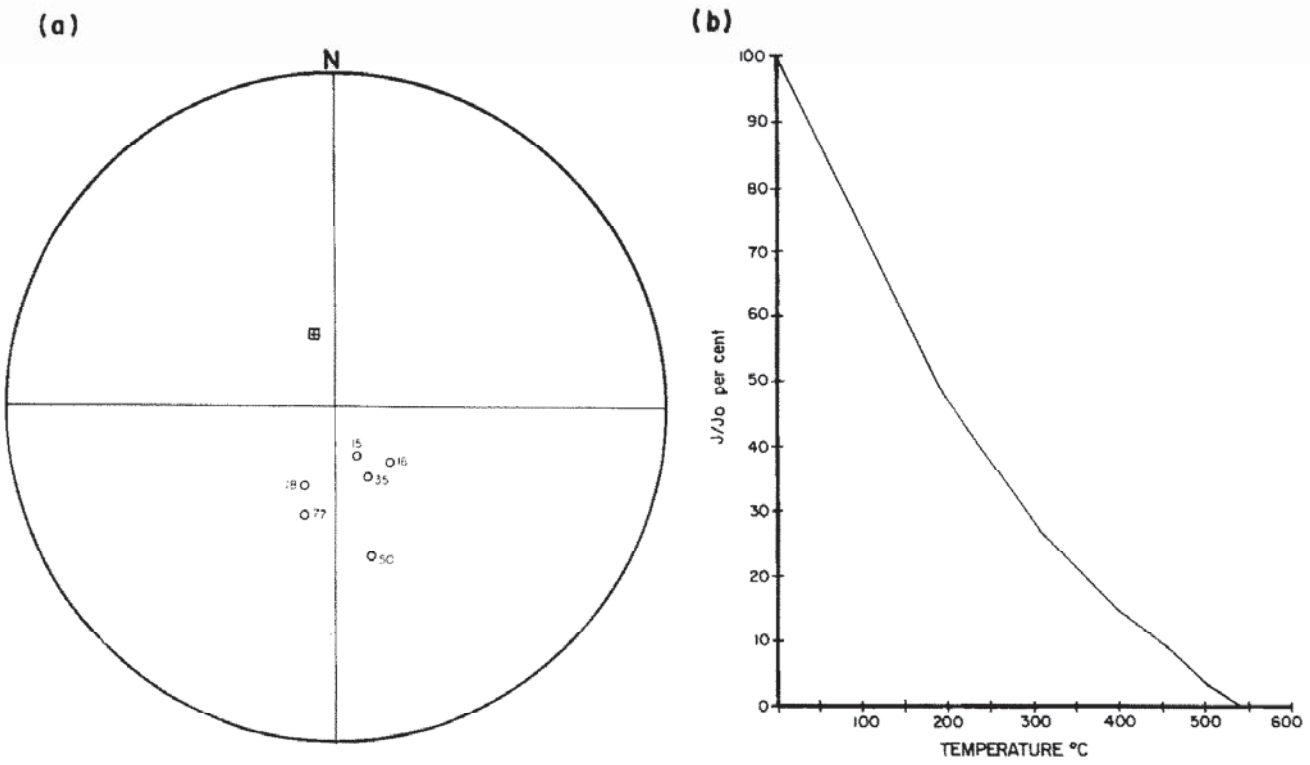


Fig. 6. a. End point vectors (see text and Fig. 5) corrected for the dip of the igneous layering. Plotting convention as in Fig. 3. b. Normalized intensity response curve to continuous thermal demagnetization of a specimen from site 16.

in its present attitude. However, the decrease of k is not significant at the 95% confidence limit and the magnetization directions used in the fold test represent one specimen from each site only. Consequently the above result should be treated with caution.

The intensity responses to thermal demagnetization of all the specimens in the upper zone are similar. Two such typical normalized intensity response curves are shown in Fig. 6b. The shape of these curves suggests a thermally well distributed blocking temperature spectrum, but the average Curie temperature obtained from all the pilot specimens is 547 °C, with a standard deviation from the mean value of 7 °C. A Curie temperature of 547 °C indicates a titanomagnetite with 0.056 mole fraction Fe_2TiO_4 (ulvöspinel), (McElhinny, 1973) in solid solution, as the source of the remanent magnetization.

Mineralogy of opaque minerals

Magnetite generally occurs in all the specimens as rather large grains of up to 1 mm in diameter, although a whole range of magnetite particles down to approximately 10 μm, are also present. The Curie temperature of the specimens indicates that this is a titanomagnetite with a cell dimension (Am) of 0.84 nM. This value is identical to the average cell dimensions of titanomagnetite in the upper zone of the layered sequence given by Molyneux (1972). Opaque exsolution lamellae in pyroxene and plagioclase crystals are rare, but some were found in specimens from sites 16 and 17, which could possibly be magnetite (see discussion Hattingh, 1986a).

The titanomagnetite consists of magnetite with a cloth-like texture formed by exsolved ulvospinel. Haggerty (1976), classified the degree of magmatic oxidation into seven stages, from C1 the lowest, to C7, the highest, which is characterized by the pseudobrookite–haematite assemblage. According to this classification the titanomagnetite of specimens from sites in the upper zone fall into the C2 and C3 oxidation stages, which are characterized by magnetite-ulvöspinel and a small number of ilmenite exsolution lamellae.

Evidence for low temperature oxidation of the titanomagnetites was found in places by the presence of maghemite at the margins of titano-mag-

netite grains. However, subsequent XRD investigation of separated titanomagnetite grains failed to confirm the presence of maghemite, which could be due to the relative scarceness of the mineral in the specimens. The XRD-data confirmed the cell dimension of 0.84 nM obtained from the average Curie temperature, for the magnetite.

The large size of the titanomagnetite grains is consistent with the MD characteristics of the AF demagnetization data at low AF values. Titanomagnetite grains of only 10 μm will, however, still be MD grains (Soffel, 1971) and although smaller titanomagnetite grains could be present in the specimens, it becomes very difficult to support evidence for SD carriers on the basis of the observed opaque minerals, apart from two sites where opaque lamellae, conceivably magnetite, have been observed in pyroxene and plagioclase crystals.

Apart from occurring as exsolution lamellae in the titanomagnetite grains, ilmenite also exists as large discrete grains in the specimens. The average cell parameters (Arh) of the coarse ilmenite grains in the upper zone of the layered sequence is 0.553 nM (Molyneux, 1972, p. 865), which indicates a mole fraction of 80% $FeTiO_3$ (McElhinny, 1973). From chemical analyses by Molyneux (1972, p. 870) the ilmenite contains, on average, less than 4 mol.% Fe_2O_3, about 80 mol.% $FeTiO_3$ and about 16 mol.% $MgTiO_2$. No haematite exsolution lamellae were observed in the ilmenite and with the previously quoted cell parameter which should result in a Curie temperature of approximately 0 °C for the ilmenite, the ilmenite in the specimens in all probability does not contribute significantly to the remanent magnetization.

Geomagnetic field reversal or self-reversal?

The mean magnetization direction of the upper zone (group BUZ1AF) is normal. It is, however, reversed with respect to the magnetization direction of the main zone in the western Bushveld (Hattingh, 1986b) and the magnetization direction of subzone B of the main zone in the eastern Bushveld (Hatttingh, 1986b). This difference in magnetic polarities can either represent a geomagnetic field reversal or a self-reversal of magnetization. Although occurrences of magnetic self-reversals can be considered rare (Merril, 1975; Tarling, 1971), with evidence overwhelmingly in favour of geomagnetic field reversals (Cox, 1969b), it is always pertinent to consider the possibility of self-reversal.

In an attempt to reproduce a self-reversal of magnetization in specimens from the upper zone, six randomly selected specimens were heated to 650 °C in presence of nitrogen gas and then left to cool in the Earth's magnetic field. The TRM's acquired by the specimens were in all instances parallel to the magnetic field in which the specimens cooled down. Further AF and thermal demagnetization of the specimens yielded no corroborative information for a possible self-reversal, or even partial self-reversal of magnetization. The evidence although overwhelmingly against self-reversal cannot however, be considered as absolute proof that a self-reversal did not take place, because the chances of reproducing a self-reversal of magnetization of specimens will depend on the degree to which experiments conducted in the laboratory conform to conditions in nature.

Specimens from the upper zone and subzone C of the main zone (Hattingh, 1986b), immediately below the upper zone in the eastern Bushveld have the same magnetic polarity (Hattingh, 1986b). If the magnetization direction of the upper zone is the result of a self-reversal, one must assume the same for subzone C of the main zone. The opaque mineralogy of the two zones are dissimilar and consequently any self-reversals of magnetization would have to have been the result of different mechanisms in each zone. Although this is not impossible, such a situation is unlikely in view of the general scarceness of self-reversals of magnetization.

According to Morgan and Briden (1981, p. 164), a geomagnetic field reversal occurred at about 1950 m.y. ago. The effect of such a reversal of the geomagnetic field with an approximate age not much different from that of the layered sequence that of the layered sequence (2049 ± 152 m.y., Hattingh, 1986b), may be manifested in the palaeomagnetic polarity of the layered sequence and can very well be the cause of the change in magnetic polarity towards the top of the layered sequence.

Summary and conclusions

Application of AF demagnetization techniques yielded a primary magnetization direction for the upper zone of the Bushveld Complex. This direction was confirmed with results from thermal demagnetization of specimens and has the following parameters:

group: BUZ1AF; $N = 7$; $D = 184.3°$; $I = 66.0°$; $\alpha_{95} = 11°$; $k = 31$;

with pole position: 16.1°S, 148.5°W;

and polar error (dp, dm): 14.7° and 18.0° respectively.

All the fold tests carried out on the palaeomagnetic data from the upper zone failed to improve group statistics significantly. This implies that the upper zone had acquired its remanent magnetization with the igneous layering in its present position. This is at variance with results from the main and critical zones (Hattingh, 1986a, 1986b, 1986c), which indicate that the lower parts of the mafic layered sequence had acquired its magnetization with the igneous layering in a horizontal position. From this, emplacement of the upper zone was the last phase in the formation of the layered sequence. Down buckling of the Bushveld floor and associated subsidence due to isostatic readjustment, took place after the critical and main zones had cooled to below its Curie temperatures. Only then did the emplacement of the upper zone take place. This chronological sequence of events is further substantiated by the palaeomagnetic pole positions of the different zones shown on the apparent polar wander path (APW) in Fig. 7.

The magnetic polarity of the upper zone is normal (N) with respect to the present geomagnetic field in southern Africa. This is opposite to the reversed (R) polarity of the main zone in the western Bushveld Complex (Hattingh, 1986c) and the polarity of subzone B of the main zone in the eastern Bushveld Complex (Hattingh, 1986b). The magnetic polarity of the critical zone (Hattingh, 1986a) is however normal. This magnetic polarity pattern is shown in Fig. 8.

In view of the above striking magnetic polarity pattern exhibited by the mafic layered sequence it is proposed that a further zonal subdivision of the

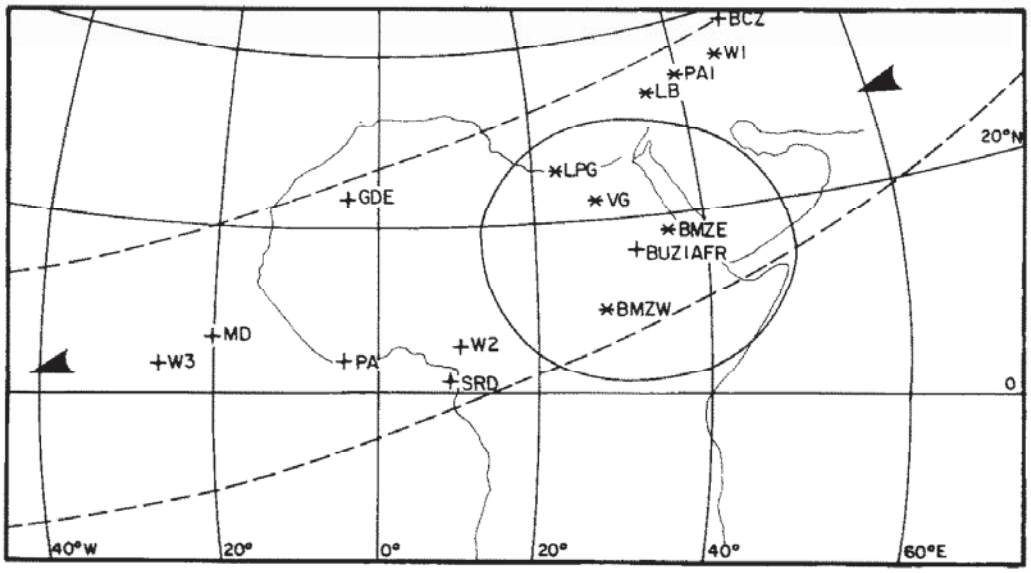


Fig. 7. APW path for Africa with selected pole positions shown for Southern Africa ca. 2070–1700 m.y. PA1 (2047 ± 11 m.y.)—Phalaborwa pyroxenite (Morgan and Briden, 1981); W1–W3—Waterberg sediments (stratigraphically arranged poles E, D and C in McElhinny, 1968); VG (2006 ± 47 m.y.)—Vredefort Granophyre (Hargraves, 1970); LPG (1980 ± 39 m.y.)—Limpopo gneisses (Morgan and Briden, 1981); LB (1840 ± 276 m.y.)—Losberg Gabbro (Hargraves, 1970); GDE (1895 ± 73 m.y.)—Great dyke extensions into Limpopo belt (Jones et al., 1975); SRD (1876 ± 68 m.y.)—Sand River dykes (Morgan, 1985); PA (2012 ± 19)—Phalaborwa syenite and dolerites (Morgan and Briden, 1981); MD (1811 ± 20 m.y.)—Mashonaland dolerites (McElhinny and Opdyke, 1964); BCZ (2054 ± 11 m.y.)—critical zone Bushveld Complex (Hattingh, 1986a); BMZE (2049 ± 152 m.y.)—main zone eastern Bushveld Complex (Hattingh, 1986b); BMZW—main zone western Bushveld Complex (Hattingh, 1986c); BUZ1AFR—upper zone Bushveld Complex, this study; * —N poles; + —south poles.

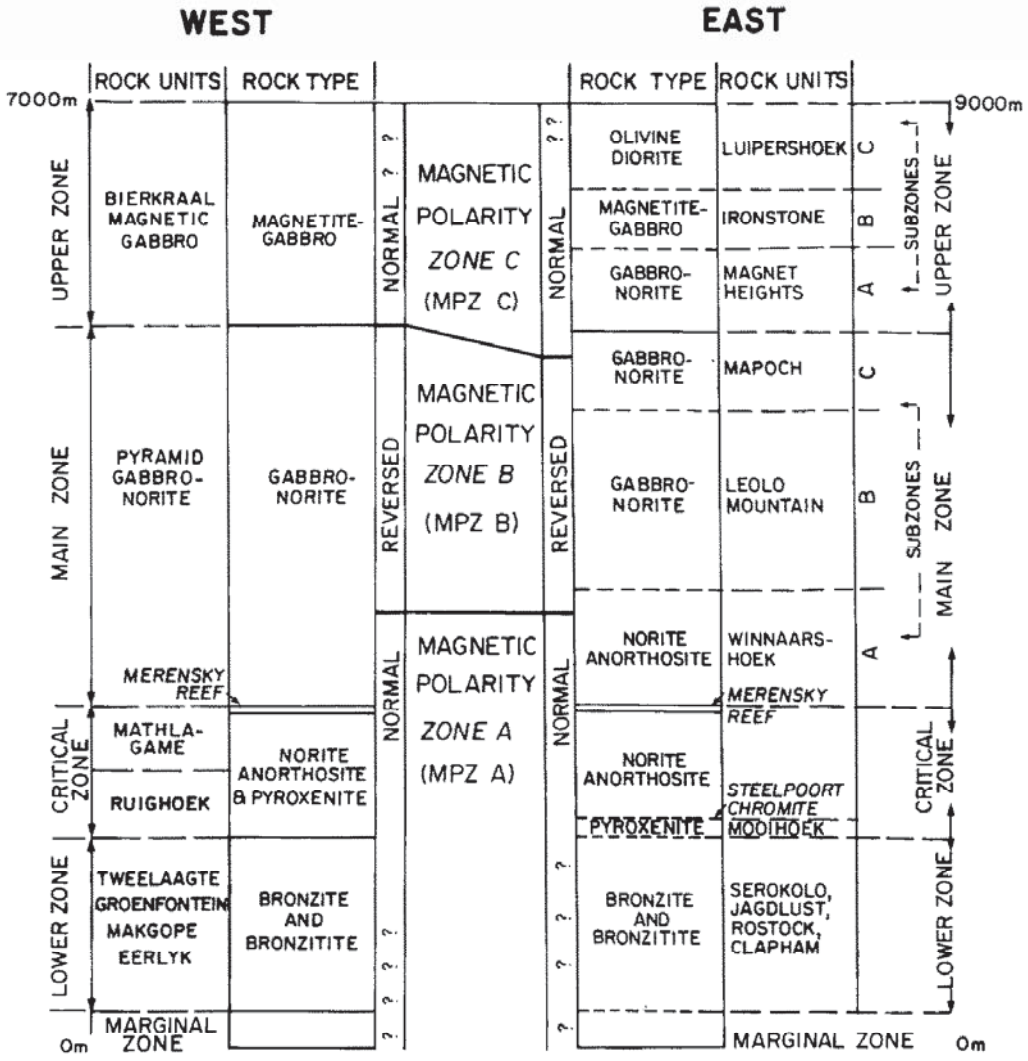


Fig. 8. Simplified lithostratigraphical column of the mafic portion of the Bushveld Complex showing magnetic polarity zones.

sequence is made on the basis of magnetic polarity. It is possible to subdivide the layered sequence into three magnetic polarity zones, namely A, B and C, (MPZA, MPZB and MPZC), as shown in Fig. 8.

Although a resemblance exists, there is no close relation between either the petrological-zonal or lithostratigraphic subdivision and the observed polarity pattern which is basically a chronostratigraphical subdivision. This apparent unconformal relationship may be pertinent to the evolution and subsequent history of the mafic layered sequence.

The polarity change between MPZA and MPZB could represent either a geomagnetic field reversal or a self-reversal of magnetization (Hattingh, 1986c). If it represents a field reversal then this event must have occurred ca. 2050 m.y. ago (Fig. 7). The polarity change between MPZB and MPZC coincide with a general polarity change on the APW ((Fig. 7). Despite the rather large oval of 95% confidence around pole BUZ1AF, this indicates a geomagnetic field reversal which occurred between 2050 and 1876 m.y. ago.

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